TopCalc

A small offline launcher for Windows, macOS, and Linux. It stays in the tray. A hotkey shows the bar. It does not start a new process.

Open What to type Calculator Graph Shell Files Build Limits

The bar searches apps and files, evaluates math, plots in 2D, runs a shell command, or opens that command in a terminal. Nothing in the app calls the network. **Search the web** opens your browser. The license is MIT.

Open it

The default shortcut is `Alt` + `Space`. `Esc` hides the bar. A left-click on the tray icon also shows it.

On Windows, Alt+Space is also the system menu of a focused window. If the hotkey does not fire, change it: tray icon, then `Ctrl` + `,` or the gear.

- Type a shortcut such as `Alt+Space` or `Ctrl+Alt+Space`.
- Or press **Record** and hit the keys. Recording suspends the current shortcut so it does not hide the window.

Settings are one JSON file:

- Windows: `%APPDATA%\com.topcalc.app\settings.json`
- macOS: `~/Library/Application Support/com.topcalc.app/settings.json`
- Linux: `~/.config/com.topcalc.app/settings.json`

```
{
  "hotkey": "Ctrl+Alt+Space",
  "searchUrl": "https://duckduckgo.com/?q={query}",
  "theme": "dark",
  "anchor": "top"
}
```

`searchUrl` must be `http` or `https`. `{query}` or `%s` is replaced with the URL-encoded text. Launch at login is not implemented yet.

Color scheme is Dark, Light, Slate, Forest, or Custom. Custom sets background, text, muted text, accent, and surface.

Position is Top, Center, Bottom, a corner, or Custom. Custom is a horizontal and vertical percent, and the bar is centered on that point. The bar moves while you change it. Save keeps it. Esc restores the last saved scheme and position.

What to type

TYPE THIS	WHAT HAPPENS
`2+2`, `sqrt(2)`, `sin(pi/2)`, `=1+1`	Evaluated locally. Enter on the result copies it.
`sin(x)`, `y = x^2`, `x = 3`, `x^2 = 4`	Opens the graph. Equations list real solutions. Several expressions are separated with `;`.
`d/dx cos x`, `int sinx dx from 0 to pi`	Derivative or integral. Indefinite integrals are graphed with C = 0.
`Ctrl` + `G`	Shows or hides the graph.
`c:`, `d:`, `c:\Users`, `/`, `~`	File search under that root.
`chrome`	Installed applications, then files when the query is at least 3 characters.
`> Get-Process`	Shell. On Windows `>` is PowerShell.
`$ ls`	On Windows, bash via Git Bash or WSL when one of those is installed.
`y = x^2`, `x = 3`, `=sin(x)`	Opens a graph window. Enter there adds another line on the same graph. A leading `=` opens the window as soon as the math is valid.
`google.com`	Enter opens that site. A leading space searches instead: `google.com` searches for `google.com`.
`sleep`	Enter puts the computer to sleep. A leading space searches instead.
anything else	**Search the web** is the Enter action when nothing local is a strong match.

On macOS and Linux, both `>` and `$` use your login shell (`$SHELL`, or `/bin/sh`).

Calculator

Supported: `+ - * / ^ **`, parentheses, and juxtaposition such as `2x`, `sinx`, and `sin2x`. `^` is right-associative, so `2^3^2` is 512. `-3^2` is `-9`. Trig is in radians. `log` and `ln` are natural log. `log10` is base 10. `sin^2(x)` means `(sin(x))^2`.

Functions take parentheses, a space, or a following factor: `arcsin(1)`, `arcsin 1`, or `arcsinx`. Leading words such as `solve`, `plot`, `what is`, and `calculate` are ignored.

Typing `sum`, `prod`, or `lim` and then a space inserts that operator with the cursor in the body. `sum_{n=1}^{10} n` is 55. `lim_{x->0} sin(x)/x` is 1. `5!` is 120.

The **∑** button opens the function keys, one group at a time: Trig, Inverse, Hyperbolic, Calculus, Functions, and Symbols. Inverse includes `arcsin`, `arccos`, and `arctan`. Calculus includes derivatives, integrals, sums, products, and limits.

Real equations in `x` are solved and graphed. `x = 3` is a vertical line. `x^2 = 4` plots both sides and lists `x = -2` and `x = 2`. Lines and quadratics are exact. Other equations are solved numerically on [−20, 20]. `y = ...` and bare expressions in `x` are plotted as `y = f(x)`.

Graph

Drag to pan. Scroll to zoom. Double-click or **Reset** restores the view. Hover shows an approximate point. Discontinuities such as `1/x` and `tan(x)` break the line.

Shell

Typing a command never runs it. The first Enter on **Run in shell** only arms it. A second Enter runs it. **Open in terminal** does not need the extra confirm.

The inline runner waits at most 8 seconds, then stops the process. On Windows the inline shell is PowerShell with `-NoProfile`. Bash is used only when Git Bash or a WSL distro is present.

Open in terminal uses Windows Terminal (`wt`) when it is on `PATH`, otherwise PowerShell. On macOS it uses Terminal.app. On Linux it tries `x-terminal-emulator` and then common terminals.

Files and apps

Applications come from the Start Menu on Windows, `/Applications`, `/System/Applications`, and `~/Applications` on macOS, and `.desktop` files on Linux.

File search is capped: about 80 ms debounce, depth 3, a few thousand directory visits, 30 results. `c:` lists the top of that drive. These directories are skipped unless you typed a path into them: `Windows`, `node_modules`, `.git`, `$Recycle.Bin`, `System Volume Information`, and on Unix `proc`, `sys`, and `dev`.

Without a path, files are searched only after 3 characters, and only under Desktop, Documents, Downloads, and a few other home folders. Enter on an app launches it. Enter on a file or folder opens it.

Build and run

You need Rust stable, Node.js 20 or newer, and a system web view: WebView2 on Windows, WebKit on macOS, WebKitGTK 4.1 on Linux.

On Debian or Ubuntu:

```
sudo apt update
sudo apt install -y libwebkit2gtk-4.1-dev build-essential libssl-dev \
  libayatana-appindicator3-dev librsvg2-dev patchelf libxdo-dev
```

```
npm install
npm run tauri dev
```

The window stays hidden until `Alt` + `Space` or a left-click on the tray icon. `npm run tauri build` makes a production build.

Limits

- One 2D explicit plotter. No 3D, and no implicit curves such as `x^2 + y^2 = 1`.
- Symbolic calculus covers common derivatives and a limited set of integrals.
- File search is capped and skips the heavy directories listed above.
- The shell confirm and the 8 second inline timeout are intentional.
- Alt+Space can lose to the Windows system menu. Change it in settings if that happens.
- Launch at login is left for later.